

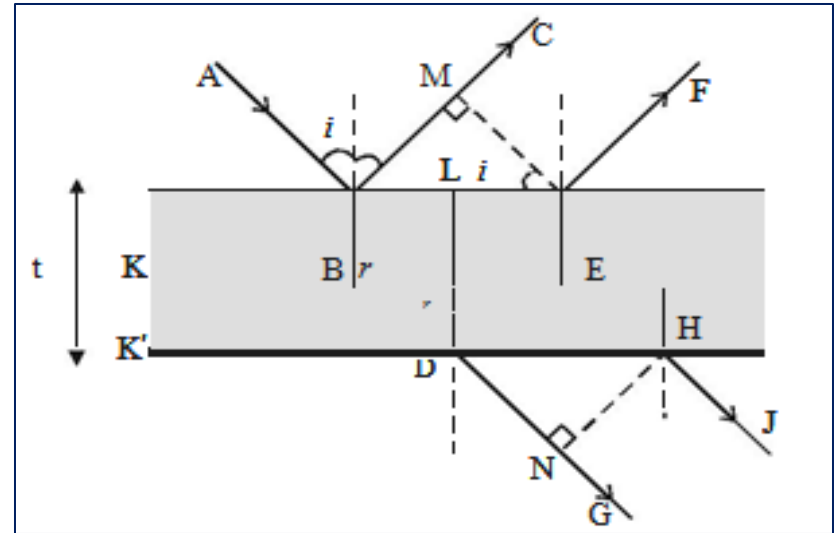
Interference in thin films

Thin film

- An optical medium is called a thin film when its thickness is about the order of 1 wavelength of light in visible region
- Thus , a film of thickness in the range $0.5\text{ }\mu\text{m}$ to $10\text{ }\mu\text{m}$ may be considered as a thin film
- A thin film may be a thin sheet of transparent material such as glass, mica, an air film enclosed between two transparent plates or a soap bubble
- When light is incident on such a film, a small part of it gets reflected from the top surface and a major part is transmitted into the film
- Again, a small part of the transmitted component is reflected back into the film by the bottom surface and the rest of it emerges out of the film
- A small portion of the light thus gets reflected partially several times in succession within the film

Interference in Thin Films

- Consider a transparent thin film of uniform thickness t and its refractive index μ bounded by two plane surfaces K and K' as shown in figure.
- A ray of monochromatic light AB incident on the surface K of the film is partly reflected along BC and partly refracted into the film along BD .
- At the point D on the surface K' , the ray of light is partly reflected along DE and partly transmitted out of the film along DG .



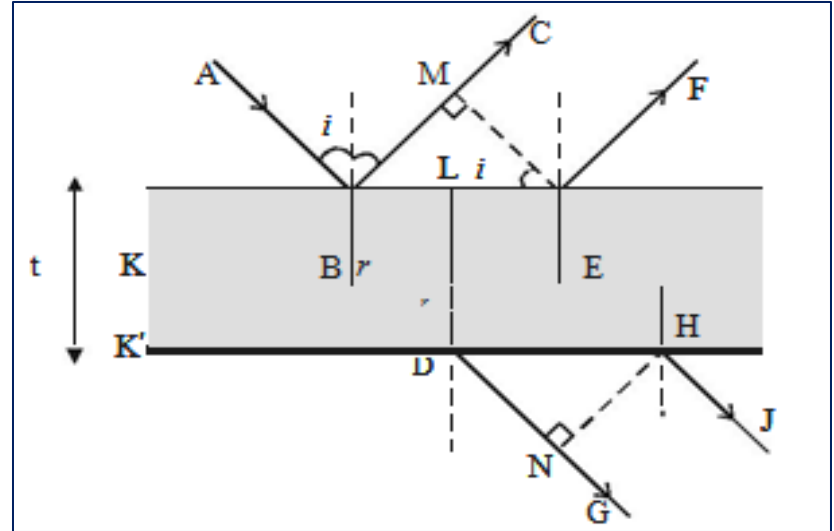
The reflected light then emerges into air along EF which is parallel to BC . The ray EH after refraction at H , finally emerges along HJ .

BC and EF are reflected rays parallel to each other and DG and HJ are transmitted rays parallel to each other. Rays BC and EF interfere and similarly the rays DG and HJ interfere.

- The waves travelling along the paths BC and BDEF are derived from a single incident wave AB
- Therefore they are coherent and can produce interference if they are made to overlap by a condensing lens or the eye

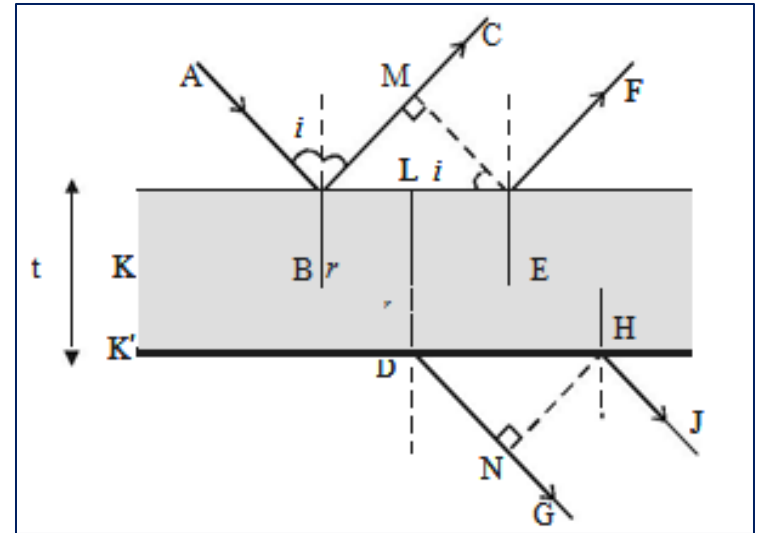
Interference due to the Reflected Beam

- EM is drawn normal to BC from E.
- Now the path difference between the waves BC and EF
- $\delta = (BD + DE)_{\text{in film}} - (BM)_{\text{in air}}$
- We know, that a distance in air is numerically equal to μ times the distance in medium
- $\delta = \mu (BD + DE) - BM$
- From the figure, it is clear that $BD = DE$
- $\delta = (2\mu \cdot BD) - BM$
- In the ΔBME , $\sin i = BM/BE$ [$\mu = \sin i / \sin r$]



- $BM = BE \sin i = BE \cdot \mu \sin r$
- $BM = \mu \cdot BE \sin r$
- In the ΔBDL , $\sin r = BL/BD = (1/2 BE)/BD$
- $BE = 2 (BD) \sin r$
- Therefore, $BM = \mu^2 (BD) \sin^2 r$

- $BM = \mu(2BD) \sin^2 r$
- $\delta = 2\mu BD - 2\mu BD \sin^2 r$
- $\delta = 2\mu BD \cos^2 r$
- In the BDL, $\cos r = DL/BD = t/BD$,
- $BD = t / \cos r$
- So, $\delta = 2 \mu (t / \cos r) \cos^2 r$
- $\delta = 2 \mu t \cos r$



Correction on account of phase change at reflection

- A ray of light travelling in air and getting reflected at the surface of a denser medium, undergoes an automatic phase change of π (or) an additional path difference of $\lambda/2$.
- Since the reflection at B is at the surface of a denser medium, there is an additional path difference $\lambda/2$.
- The effective path difference in this case, $\delta = 2\mu t \cos r + \lambda/2$

- (i) For the **constructive interference**, path difference $\delta = n\lambda$, where $n = 0, 1, 2, 3$ and the film appears bright
- $2\mu t \cos r + \lambda/2 = n\lambda$
- So, $2\mu t \cos r = (2n-1) \lambda/2$
- (ii) For the **destructive interference**, path difference
- $\delta = (2n+1) \lambda/2$
- where $n = 0, 1, 2, 3 \dots$ and the film appears dark.

- $2\mu t \cos r + \lambda/2 = (2n+1)\lambda/2$
- $2\mu t \cos r = n\lambda$
- If light is incident normally $i = 0$ and hence $r = 0$.
- Therefore the condition for bright fringe is $2\mu t = (2n-1)\lambda/2$ and for dark fringe is $2\mu t = n\lambda$.

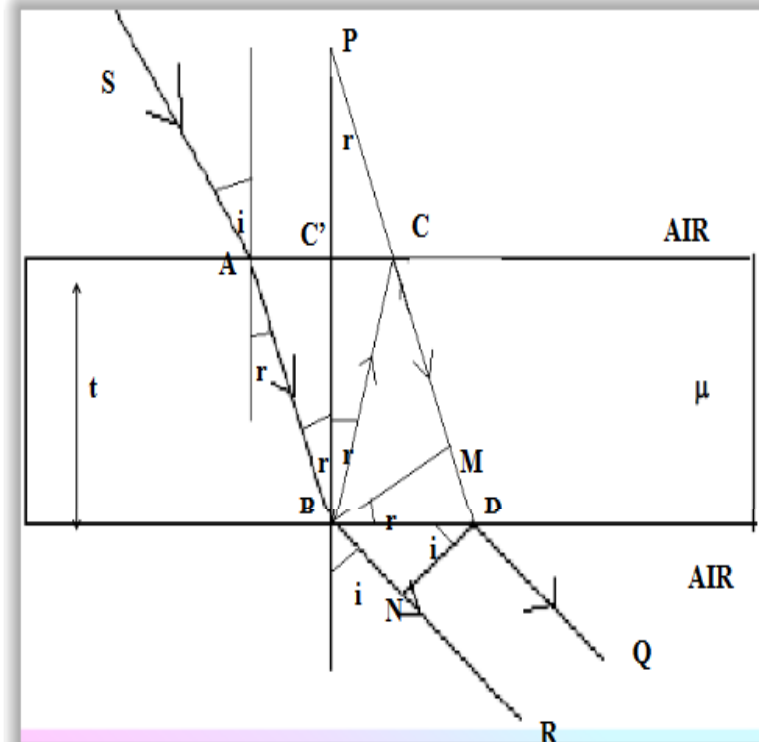
Interference due to transmitted light in thin films

Here at B reflection takes place at the rarer Medium .so no phase change occurs.

BM normal to CD and DN normal to BR.

So optical path difference between DQ and BR is

$$x = \mu(BC+CD) - BN$$



$$n_1 \sin i = n_2 \sin r$$

$$\mu \sin r = \sin i$$

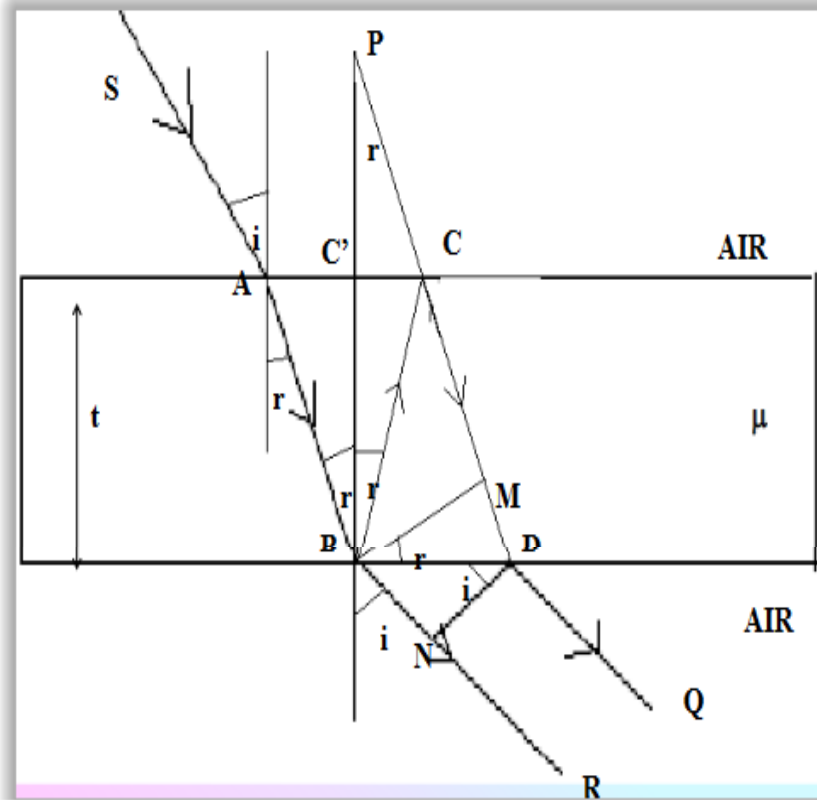
$$\mu = \sin i / \sin r$$

$$\text{In } \triangle BND, \quad \sin i = BN/BD$$

$$\text{In } \triangle BMD, \quad \sin r = MD/BD$$

$$\mu = \sin i / \sin r = BN/MD$$

$$\Rightarrow BN = \mu MD$$



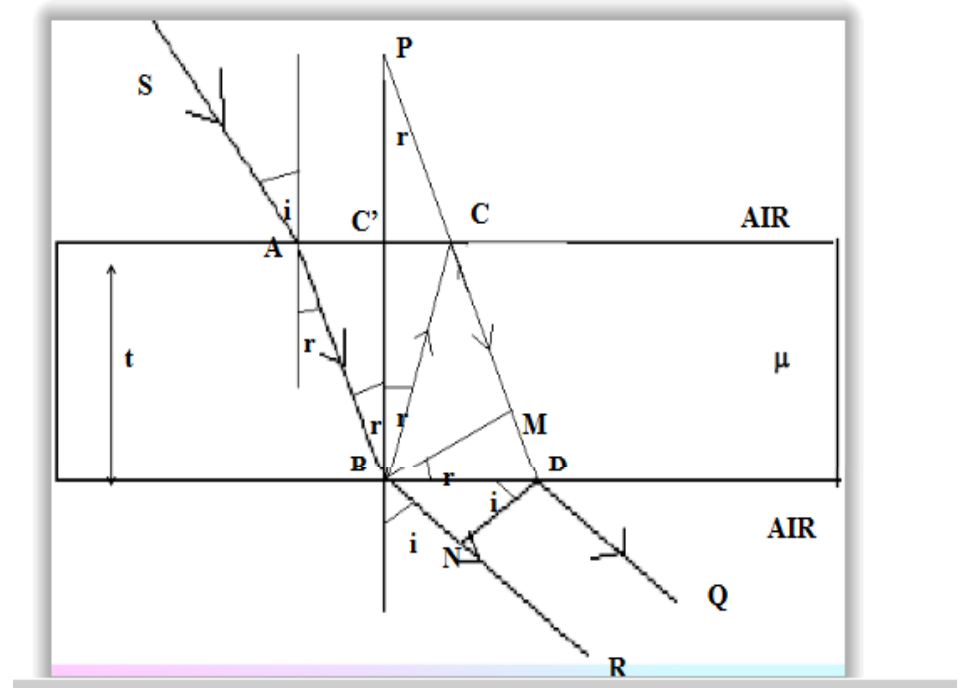
Since $\angle CPB = \angle CBP = r$
and $BC = CP = CD$

$$\begin{aligned} x &= \mu(BC + CD) - BN \\ &= \mu(DP) - \mu(MD) \\ &= \mu(DP - MD) \\ &= \mu(MP) \end{aligned}$$

In $\triangle BPM$ $\cos r = MP/BP$

So $MP = BP \cos r$

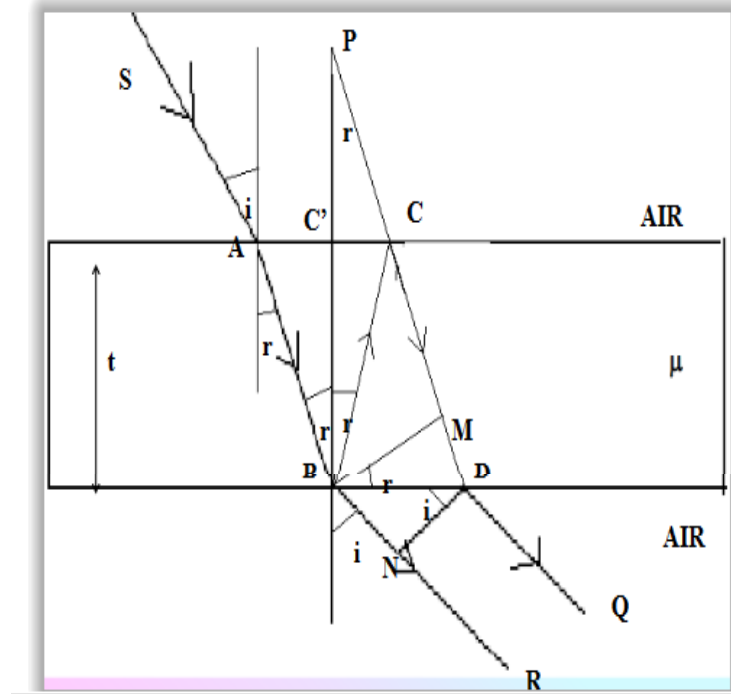
$= 2t \cos r$ since $BP = 2t$



Path difference , $x = 2 \mu t \cos r$

Conditions for bright and dark fringes

- The path difference between the transmitted rays BR and DQ is, in a similar way, $\delta = 2\mu t \cos r$.
- In this case there is no additional path difference introduced because both reflections at the point B and C take place backed by rarer medium
- Hence, condition for brightness is $2\mu t \cos r = n\lambda$ and condition for darkness is $2\mu t \cos r = (2n - 1)\lambda/2$.



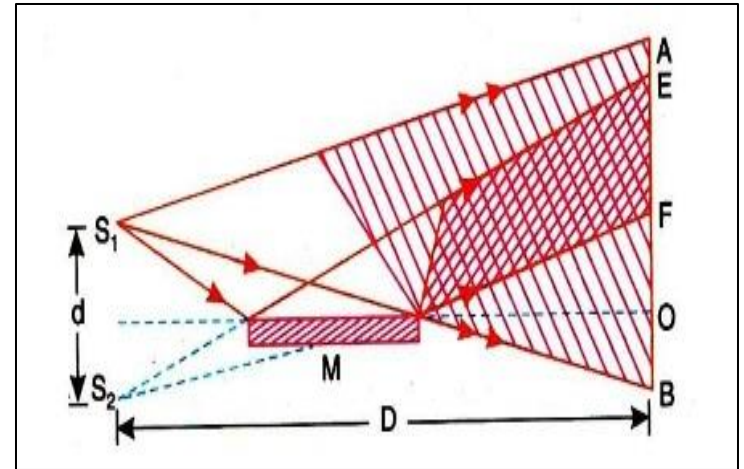
Applications

- Thin films are used commercially in anti-reflection coatings, mirrors, and optical filters.
- They can be engineered to control the amount of light reflected or transmitted at a surface for a given wavelength.
- A Fabry–Pérot etalon takes advantage of thin film interference to selectively choose which wavelengths of light are allowed to transmit through the device.
- These films are created through deposition processes in which material is added to a substrate in a controlled manner.
- Methods include chemical vapor deposition and various physical vapor deposition techniques.

Lloyd's Single Mirror:

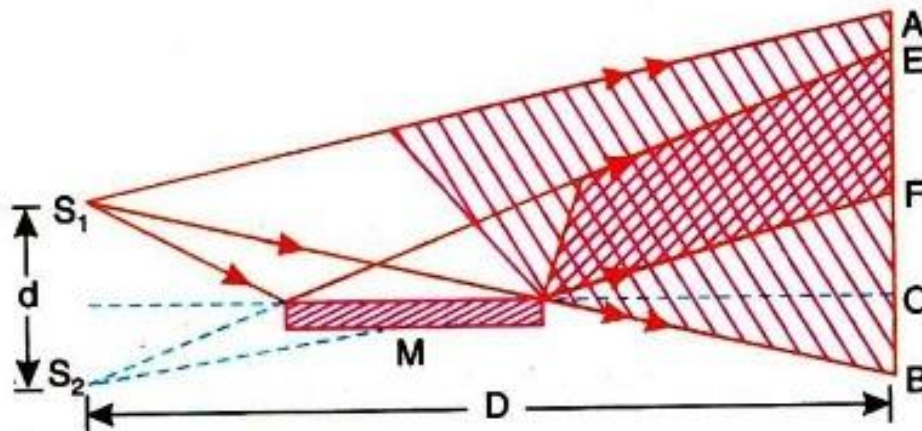
- In 1834, Lloyd devised an interesting method of producing interference, using a single mirror and using almost grazing incidence.
- The Lloyd's mirror consists of a plane mirror about 30cm in length and 6 to 8 cm in breadth
- It is polished on the front surface and blacked at the back to avoid multiple reflections

- A cylindrical wave front coming from a narrow slit S_1 falls on the mirror which reflects a portion of the incident wavefront, giving rise to a virtual image of the slit S_2
- Another portion of the wave front proceeds directly from the slit S_1 to the screen
- The slits S_1 and S_2 act as two coherent sources

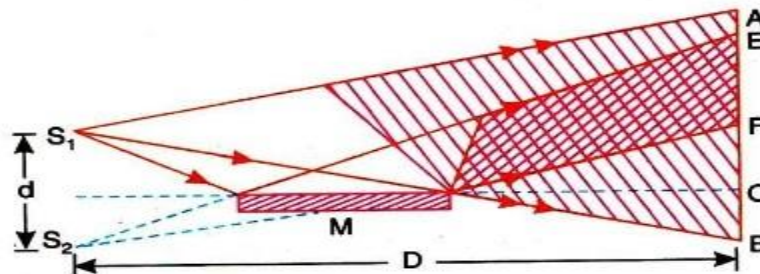


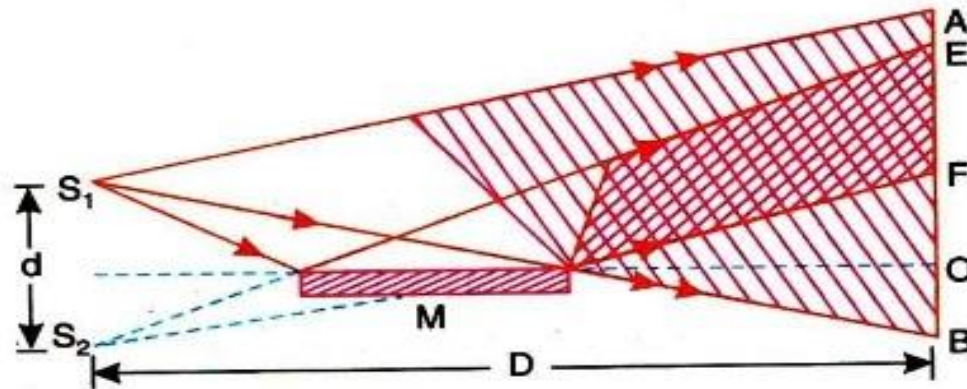
Interference between direct and reflected waves occurs within the region of overlapping of the two beams and fringes are produced on the screen placed at a distance D from S_1 in the shaded portion EF

- The point O is equidistant from S_1 and S_2
- Therefore, central (zero-order) fringe is expected to lie at O (the perpendicular bisector of S_1S_2) and it is also expected to be bright
- However, it is not usually seen since the point O lies outside the region of interference (only the direct light and not the reflected light reaches O)



- By moving the screen nearer to the mirror such that it comes into contact with the mirror, the point O can be just brought into the region of interference
- With white light the central fringe at O is expected to be white but in practice it is dark
- The occurrence of dark fringe can be understood taking into the consideration of the phase change of π that light suffers when reflected from the mirror
- The phase change leads to a path difference of $\lambda/2$ and hence destructive interference occurs there.





- **Determination of Wavelength:**
- The fringe width is given by $\beta = \frac{\lambda D}{d}$
- Measuring β , D and d the wavelength λ can be determined.

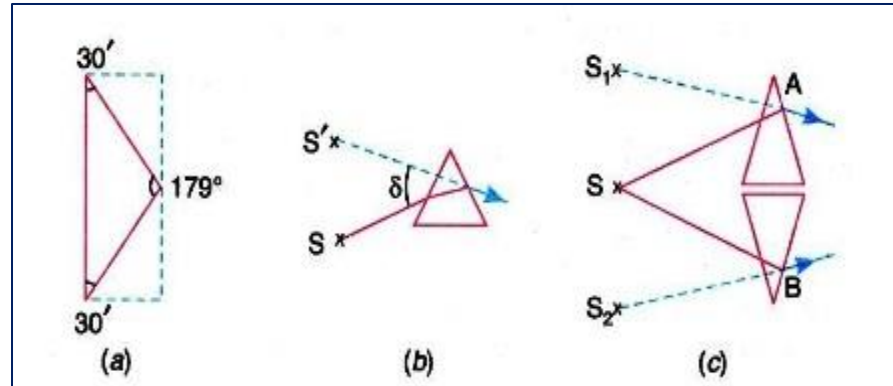
Applications of Lloyd's mirror

Interference lithography

- The most common application of Lloyd's mirror is in UV photolithography and nano patterning. Lloyd's mirror has important advantages over double-slit interferometers.
- If one wishes to create a series of closely spaced interference fringes using a double-slit interferometer, the spacing d between the slits must be increased.
- Increasing the slit spacing, however, requires that the input beam be broadened to cover both slits.
- This results in a large loss of power.
- In contrast, increasing d in the Lloyd's mirror technique does not result in power loss, since the second "slit" is just the reflected virtual image of the source.

- Hence, Lloyd's mirror enables the generation of finely detailed interference patterns of sufficient brightness for applications such as photolithography.
- Typical uses of Lloyd's mirror photolithography would include fabrication of diffraction gratings for surface encoders and patterning the surfaces of medical implants for improved bio-functionality
- Lithography: the process of printing from a plane surface (such as a smooth stone or metal plate) on which the image to be printed is ink-receptive and the blank area ink-repellent
- Photolithography: lithography using plates made photographically.

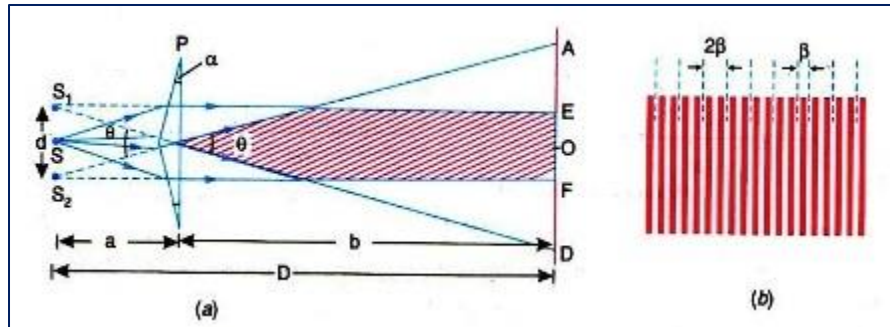
Fresnel Biprism:



- Fresnel used a biprism to show interference phenomenon.
- The biprism consists of two prisms of very small refracting angles joined base to base.
- In practice, a thin glass plate is taken and one of its faces is ground and polished till a prism is formed with an angle of about 179° and two side angles of the order of $30'$.

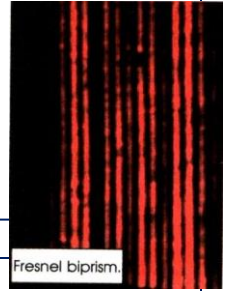
- When a light ray is incident on an ordinary prism, the ray is bent through an angle called the **angle of deviation**.
- As a result, the ray emerging out of the prism appears to have emanated from a source S' located at a small distance above the real source, as shown in figure 1b.
- The prism produced a **virtual image** of the source.
- A biprism creates two virtual sources S_1 and S_2 as shown in figure 1C. These two virtual sources are images of the same source S produced by refraction and are hence coherent.

Experimental Arrangement:



- The biprism is mounted suitably on an optical bench
- A single cylindrical wavefront incident on both prisms.
- The top portion of wavefront is refracted downward and appear to have emanated from the virtual image S_1
- The lower segment, falling on the lower part of the biprism, is refracted upward and appears to have emanated from the virtual source S_2
- The virtual sources S_1 and S_2 are coherent, and hence the light waves are in a position to interfere in the region beyond the bi-prism
- If a screen is held, interference fringes are seen
- In order to observe fringes, a micrometer eyepiece is used

Experimental Fringes



Theory:

- As point O is equidistant from S1 and S2, the central bright fringe of maximum intensity occurs there
- On both sides of O, alternate bright and dark fringes are produced
- The width of the dark or bright fringe is given by

$$\beta = \frac{\lambda D}{d} \quad (1)$$

- Where $D = (a+b)$ is the distance of the source from the eyepiece

Comparison between the fringes produced by biprism and Lloyd's mirror:

Sr No	Fresnel Biprism	Lloyd's mirror
1	The complete set of fringes is obtained	A few fringes on one side of the central fringe are observed and the central fringe being itself invisible
2	The central fringe is bright	The central fringe is dark
3	The central fringe is less sharp	The central fringe is sharp

Optical measurement

- The output of a Lloyd's mirror was analyzed with a CCD photodiode array to produce a compact, broad range, high accuracy Fourier transform wave meter that could be used to analyze the spectral output of pulsed lasers.

Radio astronomy

- In the late 1940s and early 1950s, CSIRO scientists used a technique based on Lloyd's mirror to make accurate measurements of the position of various galactic radio sources from coastal sites in New Zealand and Australia.
- the technique was to observe the sources combining direct and reflected rays from high cliffs overlooking the sea.
- After correcting for atmospheric refraction, these observations allowed the paths of the sources above the horizon to be plotted and their celestial coordinates to be determined.

Underwater acoustics

- An acoustic source just below the water surface generates constructive and destructive interference between the direct path and reflected paths. This can have a major impact on sonar operations.
- The Lloyd mirror effect has been implicated as having an important role in explaining why marine animals such as manatees and whales have been repeatedly hit by boats and ships.
- Interference due to Lloyd's mirror results in low frequency propeller sounds not being discernible near the surface, where most accidents occur.
- This is because at the surface, sound reflections are nearly 180 degrees out of phase with the incident waves.
- Combined with spreading and acoustic shadowing effects, the result is that the marine animal is unable to hear an approaching vessel before it has been run over or entrapped by the hydrodynamic forces of the vessel's passage

Thank you